WESTERN MICHIGAN UNIVERSITY

Collegue of Engineering and Applied Sciences

Course Name: Lab. Intro. To Microprocessors

Course Number: ECE 2510

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Lab #1

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Task 1: The C code

ECE2510_Lab1_code

```c
#include "stm32l476xx.h"

void delay(int count);

int main(void) {
    /* 1. Enable Clock for GPIO Port A */
    RCC->AHB2ENR |= RCC_AHB2ENR_GPIOAEN;

    /* 2. Configure PA5 as Output */
    // Each pin in MODER uses 2 bits. PA5 uses bits 10 and 11.
    // Logic: Clear bits 10 and 11, then set bit 10 to '1' for Output mode (01).
    GPIOA->MODER &= ~(3 << 10); // Clear bits 10 and 11 (3 is 11 in binary)
    GPIOA->MODER |= (1 << 10);  // Set bit 10 to 1

    while (1) {
        /* 3. Toggle the LED using BSRR for safety */
        GPIOA->BSRR = (1 << 5);      // Set PA5 High
        delay(500000);

        GPIOA->BSRR = (1 << (5 + 16)); // Reset PA5 Low (Bit 21)
        delay(500000);
    }
}

void delay(int count) {
    volatile int i;
    for (i = 0; i < count; i++);
}
```

Figure 1: Lab 1 code to be pasted in our main.c file

The group followed the guidance of the lab instructor and created a separate folder on the lab computer, there, we separated the lab 1 work in a “lab1” folder. On Keil uVision we created a new project on the lab1 folder and started our lab.

The group followed the lab 1 instructions and copied the lab 1 code from the elearning platform into a newly created main.c file in our project.

Task 2: Simulating the Program

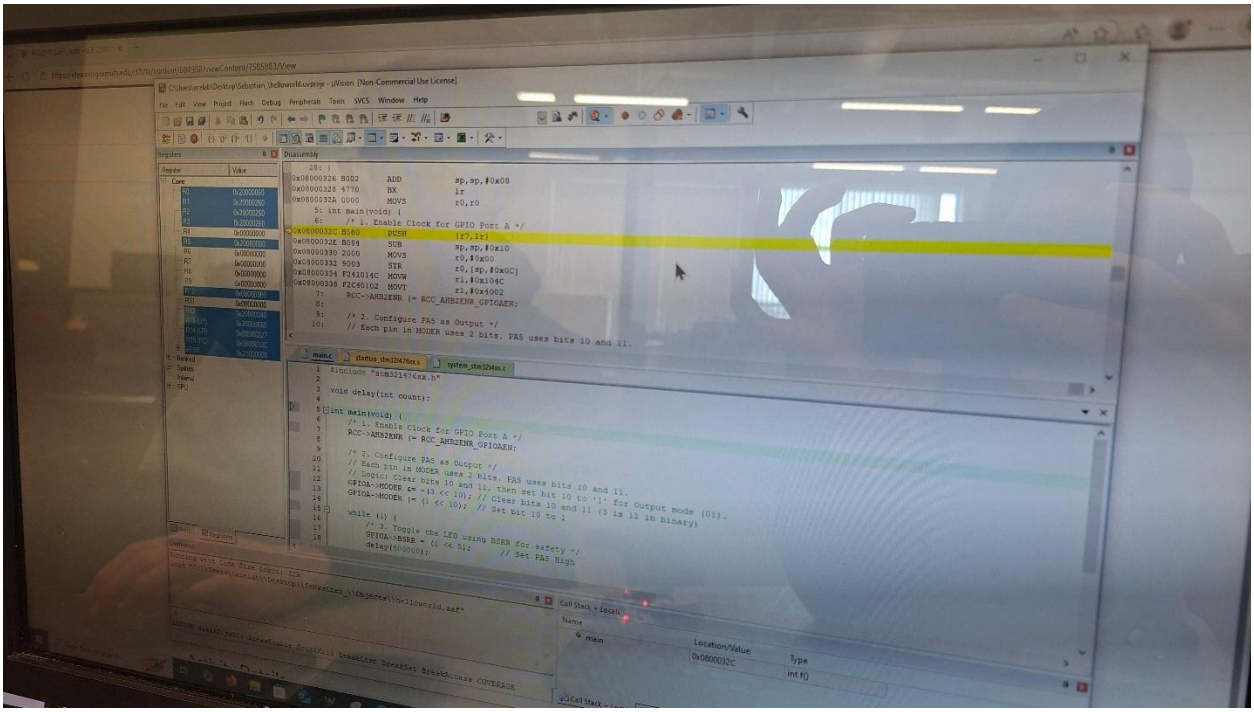

Figure 2: Lab 1 code to blink an LED running after compilation

Task 3: Running the program on the Nucleo Board

For task 3 the group followed the statements, to run the code on the nucleo board.

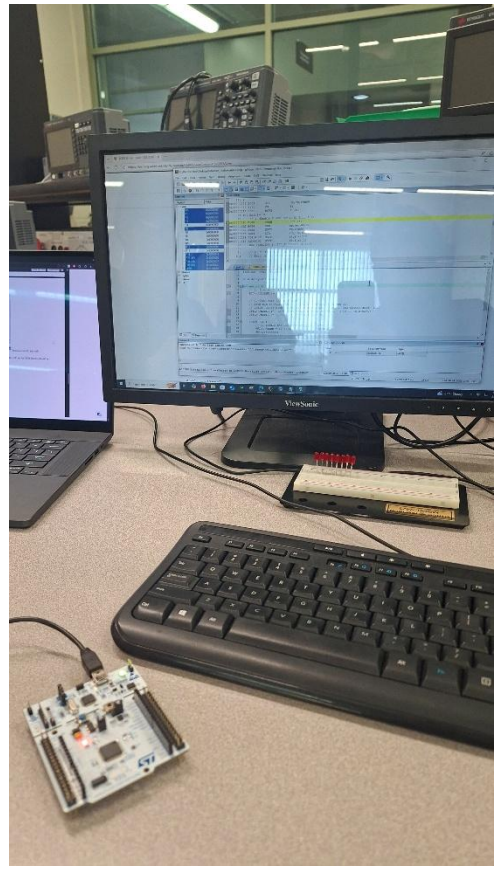

Figure 3: Nucleo Board running the LED blinking code

The code first imports the board header files wich contains the function declarations , constants and macros with: #include "stm32l476xx.h"

The code declares a function delay:

void delay(int count);

that is used to wait for some time while the program counts to one number.

Then, in the main function the program first enables clock for GPIO Port A with

RCC->AHB2ENR |= RCC_AHB2ENR_GPIOAEN;

Then, Configure PA5 as Output

Each pin in MODER uses 2 bits. PA5 uses bits 10 and 11.

Logic: Clear bits 10 and 11, then set bit 10 to '1' for Output mode (01).

Clear bits 10 and 11 (3 is 11 in binary)

GPIOA->MODER &= ~(3 << 10);

Set bit 10 to 1

GPIOA->MODER |= (1 << 10);

Then it toggles on and off the LED with an infinite loop:

```
while (1) {
    GPIOA->BSRR = (1 << 5);      // Set PA5 High
    delay(500000);
    GPIOA->BSRR = (1 << (5 + 16)); // Reset PA5 Low (Bit 21)
    delay(500000);
}
```

Conclusion

First, we created the project on a separate folder, copied the code form the elearning, whe compiled it and ran it. We were able to see the registers and memory allocation of variables, then, our group connected the Nucleo Board and uploaded the code to the board and were able to see the blinking LED.

On task 3, when my partner and me where loading the program to the Board, we were facing an issue with the LED. The code compiled and ran on the simulator but when we tried running it on the board the LED was not blinking, we were sure we loaded it correctly because the board had a blinking LED program from a previous lab session, when we loaded our program, the LED stopped blinking, we resorted to clicking the Reset push button on the board and our code started working.